

Skill Session 1

Task 1

```
j_rakshitha@RAKSHITHA:~$ gcc --version
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
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git version 2.53.0
GNU Make 4.4.1
Built for x86_64-pc-linux-gnu
Copyright (C) 1988-2023 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <https://gnu.org/licenses/gpl.html>
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
j_rakshitha@RAKSHITHA:~$ mkdir -p ~/ossproject
j_rakshitha@RAKSHITHA:~$ cd ~/ossproject
j_rakshitha@RAKSHITHA:~/ossproject$ mkdir src include build
j_rakshitha@RAKSHITHA:~/ossproject$ touch src/main.c
touch Makefile
j_rakshitha@RAKSHITHA:~/ossproject$ ls
Makefile  build  include  src
j_rakshitha@RAKSHITHA:~/ossproject$ tree
.
├── Makefile
├── build
├── include
└── src
    └── main.c

4 directories, 2 files
j_rakshitha@RAKSHITHA:~/ossproject$ nano src/main.c
j_rakshitha@RAKSHITHA:~/ossproject$ cat src/main.c
#include <stdio.h>

int main() {
    printf("OSSP Project Started Successfully!\n");
    return 0;
}
j_rakshitha@RAKSHITHA:~/ossproject$ nano Makefile
j_rakshitha@RAKSHITHA:~/ossproject$ cat Makefile
```

```

j_rakshitha@RAKSHITHA:~/ossp_project$ cat Makefile
CC = gcc
CFLAGS = -Wall -Wextra -g

TARGET = build/main
SRC = src/main.c

all:
    $(CC) $(CFLAGS) $(SRC) -o $(TARGET)

clean:
    rm -f $(TARGET)

run: all
    ./${TARGET}
j_rakshitha@RAKSHITHA:~/ossp_project$ make
gcc -Wall -Wextra -g src/main.c -o build/main
j_rakshitha@RAKSHITHA:~/ossp_project$ ls build
main
j_rakshitha@RAKSHITHA:~/ossp_project$ ./build/main
OSSP Project Started Successfully!
j_rakshitha@RAKSHITHA:~/ossp_project$ |

```

Task 2

```

j_rakshitha@RAKSHITHA:~/ossp_project$ nano src/process_demo.c
j_rakshitha@RAKSHITHA:~/ossp_project$ gcc -Wall -Wextra -g src/process_demo.c -o build/process_demo
j_rakshitha@RAKSHITHA:~/ossp_project$ ./build/process_demo
Parent Process Started
Parent PID: 1228

Parent waiting for child...

Child Process Created
Child PID: 1229
Parent PID: 1228
Child is executing ls command...
total 16
-rw-r--r-- 1 j_rakshitha j_rakshitha 164 Aug 17 05:32 Makefile
drwxr-xr-x 2 j_rakshitha j_rakshitha 4096 Aug 17 05:39 build
drwxr-xr-x 2 j_rakshitha j_rakshitha 4096 Aug 17 05:28 include
drwxr-xr-x 2 j_rakshitha j_rakshitha 4096 Aug 17 05:39 src
Child process completed.
Parent process exiting.
j_rakshitha@RAKSHITHA:~/ossp_project$ pstree -p
systemd(1)─agetty(238)
            └─chronyd-starter(108)─chronyd(272)─chronyd(282)
            └─cron(109)
            └─dbus-daemon(110)
            └─init-systemd(Ub(2))─SessionLeader(725)─Relay(733)(726)─ba+
                                └─SessionLeader(939)─Relay(941)(940)─ba+
                                └─init(7)─{init}(8)
                                └─login(329)─bash(413)
                                └─{init-systemd(Ub)}(9)
            └─networkd-dispat(157)
            └─rsyslogd(201)─{rsyslogd}(219)
                            └─{rsyslogd}(220)
                            └─{rsyslogd}(221)
            └─systemd(383)─(sd-pam)(385)
            └─systemd-journal(47)
            └─systemd-logind(168)
            └─systemd-resolve(82)
            └─systemd-udev(92)
            └─unattended-upgr(242)─{unattended-upgr}(295)
j_rakshitha@RAKSHITHA:~/ossp_project$ |

```

Skill Session 2

Task 1

```
j_rakshitha@RAKSHITHA:~/ossp_project$ ~/ossp_project
-bash: /home/j_rakshitha/ossp_project: Is a directory
j_rakshitha@RAKSHITHA:~/ossp_project$ nano src/myshell.c
j_rakshitha@RAKSHITHA:~/ossp_project$ gcc -Wall -Wextra -g src/myshell.c -o build/myshell
j_rakshitha@RAKSHITHA:~/ossp_project$ ./build/myshell
Mini Shell Started
Type 'exit' to quit.

myshell> exit
Exiting shell...
j_rakshitha@RAKSHITHA:~/ossp_project$ |
```

Task 2

```
j_rakshitha@RAKSHITHA:~/ossp_project$ nano src/keyboard_shell.c
j_rakshitha@RAKSHITHA:~/ossp_project$ gcc -Wall -Wextra -g src/keyboard_shell.c -o build/keyboard_shell
j_rakshitha@RAKSHITHA:~/ossp_project$ ./build/keyboard_shell
Keyboard Input Shell
Type commands and press Enter.
Use Backspace to edit.
Type 'exit' to quit.

myshell> hey there
Command entered: hey there
myshell> exit
Exiting shell...
j_rakshitha@RAKSHITHA:~/ossp_project$ tree
.
├── Makefile
├── build
│   ├── keyboard_shell
│   ├── main
│   ├── myshell
│   └── process_demo
├── include
└── src
    ├── keyboard_shell.c
    ├── main.c
    ├── myshell.c
    └── process_demo.c

4 directories, 9 files
j_rakshitha@RAKSHITHA:~/ossp_project$ |
```